
```
Package xAct`xPerm` version 1.2.3, {2015, 8, 23}
```

```
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```

```
Connecting to external linux executable...
```

```
Connection established.
```

```
Package xAct`xTensor` version 1.2.0, {2021, 10, 17}
```

```
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```

```
Package xAct`xPlain` version 0.0.0–developer, {2025, 5, 8}
```

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```

Particle spectroscopy

```
Get["xAct`PSALTer`"];
```

```
Package xAct`SymManipulator` version 0.9.5, {2021, 9, 14}
```

```
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```

Package xAct`xPert` version 1.0.6, {2018, 2, 28}

Copyright (C) 2005–2020, David Brizuela, Jose M. Martin–Garcia
and Guillermo A. Mena Marugan, under the General Public License.

** Variable \$CovDFormat changed from Prefix to Postfix

** Option AllowUpperDerivatives of ContractMetric changed from False to True

** Option MetricOn of MakeRule changed from None to All

** Option ContractMetrics of MakeRule changed from False to True

Package xAct`Invar` version 2.0.5, {2013, 7, 1}

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D. Yllanes and R. Portugal, under the General Public License.

** DefConstantSymbol: Defining constant symbol sigma.

** DefConstantSymbol: Defining constant symbol dim.

** Option CurvatureRelations of DefCovD changed from True to False

** Variable \$CommuteCovDsOnScalars changed from True to False

Package xAct`xCoba` version 0.8.6, {2021, 2, 28}

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Jose M. Martin–Garcia, under the General Public License.

Package xAct`xTras` version 1.4.2, {2014, 10, 30}

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** Variable \$CovDFormat changed from Postfix to Prefix

** Option CurvatureRelations of DefCovD changed from False to True

Package xAct`PSALter` version 2.0.1, {2025, 6, 5}

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Claire Rigouzzo and Haochen Tu, under the General Public License.



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```
DefField[kNS[-a, -b, -c], Antisymmetric[{-b, -c}],
  PrintAs → "K", PrintSourceAs → "J"];
```

Fundamental field	Symmetries
$\mathcal{K}_{\alpha\beta\chi}$	Symmetry[3, $\mathcal{K}^{\bullet 1\bullet 2\bullet 3}$, { $\bullet 1 \rightarrow -a$, $\bullet 2 \rightarrow -b$, $\bullet 3 \rightarrow -c$ }, StrongGenSet[{2, 3}, GenSet[-
SO(3) irrep	Symmetries
$\mathcal{K}_{0^+}^{\#1}$	Symmetry[0, $\mathcal{K}_{0^+}^{\#1}$, {}, StrongGenSet[{}], GenSet[{}]]
$\mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi}$	Symmetry[3, $\mathcal{K}_{0^+}^{\#1\bullet 1\bullet 2\bullet 3}$, { $\bullet 1 \rightarrow -a$, $\bullet 2 \rightarrow -b$, $\bullet 3 \rightarrow -c$ }, StrongGenSet[{1, 2, 3}, GenSet[-
$\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$	Symmetry[2, $\mathcal{K}_{1^+}^{\#1\bullet 1\bullet 2}$, { $\bullet 1 \rightarrow -a$, $\bullet 2 \rightarrow -b$ }, StrongGenSet[{1, 2}, GenSet[-(1,2)]]]
$\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$	Symmetry[2, $\mathcal{K}_{1^+}^{\#2\bullet 1\bullet 2}$, { $\bullet 1 \rightarrow -a$, $\bullet 2 \rightarrow -b$ }, StrongGenSet[{1, 2}, GenSet[-(1,2)]]]
$\mathcal{K}_{1^+}^{\#1}{}_{\alpha}$	Symmetry[1, $\mathcal{K}_{1^+}^{\#1\bullet 1}$, { $\bullet 1 \rightarrow -a$ }, StrongGenSet[{}], GenSet[{}]]
$\mathcal{K}_{1^+}^{\#2}{}_{\alpha}$	Symmetry[1, $\mathcal{K}_{1^+}^{\#2\bullet 1}$, { $\bullet 1 \rightarrow -a$ }, StrongGenSet[{}], GenSet[{}]]
$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$	Symmetry[2, $\mathcal{K}_{2^+}^{\#1\bullet 1\bullet 2}$, { $\bullet 1 \rightarrow -a$, $\bullet 2 \rightarrow -b$ }, StrongGenSet[{1, 2}, GenSet[(1,2)]]]
$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$	Symmetry[3, $\mathcal{K}_{2^+}^{\#1\bullet 1\bullet 2\bullet 3}$, { $\bullet 1 \rightarrow -a$, $\bullet 2 \rightarrow -b$, $\bullet 3 \rightarrow -c$ }, StrongGenSet[{1, 2}, GenSet[-

```
DefConstantSymbol[K1, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{K}\)\(\*OverscriptBox[\(K\),\((4)\)]\)\,\(1\)]\)\)";
```

```
DefConstantSymbol[K2, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{K}\)\(\*OverscriptBox[\(K\),\((4)\)]\)\,\(2\)]\)\)";
```

```
DefConstantSymbol[K3, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{K}\)\(\*OverscriptBox[\(K\),\((4)\)]\)\,\(3\)]\)\)";
```

```
DefConstantSymbol[K4, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{K}\)\(\*OverscriptBox[\(K\),\((4)\)]\)\,\(4\)]\)\)";
```

```
DefConstantSymbol[K5, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{K}\)\(\*OverscriptBox[\(K\),\((4)\)]\)\,\(5\)]\)\)";
```

```
DefConstantSymbol[K6, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{K}\)\(\*OverscriptBox[\(K\),\((4)\)]\)\,\(6\)]\)\)";
```

```
DefConstantSymbol[K7, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{K}\)\(\*OverscriptBox[\(K\),\((4)\)]\)\,\(7\)]\)\)";
```

```
DefConstantSymbol[K8, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{K}\)\(\*OverscriptBox[\(K\),\((4)\)]\)\,\(8\)]\)\)";
```

```
DefConstantSymbol[K9, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{K}\)\(\*OverscriptBox[\(K\),\((4)\)]\)\,\(9\)]\)\)";
```

```
DefConstantSymbol[K10, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{K}\)\(\*OverscriptBox[\(K\),\((4)\)]\)\,\(10\)]\)\)";
```

```
DefConstantSymbol[K11, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{K}\)\(\*OverscriptBox[\(K\),\((4)\)]\)\,\(11\)]\)\)";
```

```
DefConstantSymbol[K12, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{K}\)\(\*OverscriptBox[\(K\),\((4)\)]\)\,\(12\)]\)\)";
```

```
DefConstantSymbol[K13, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{K}\),\(\*OverscriptBox[\(\kappa\),\((4)\)]\)],\((13)\)]\)");
```

```
DefConstantSymbol[K14, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{K}\),\(\*OverscriptBox[\(\kappa\),\((4)\)]\)],\((14)\)]\)");
```

```
DefConstantSymbol[K15, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{K}\),\(\*OverscriptBox[\(\kappa\),\((4)\)]\)],\((15)\)]\)");
```

```
DefConstantSymbol[K16, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{K}\),\(\*OverscriptBox[\(\kappa\),\((4)\)]\)],\((16)\)]\)");
```

```
DefConstantSymbol[M1, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{M}\),\(\*OverscriptBox[\(\kappa\),\((2)\)]\)],\((1)\)]\)");
```

```
DefConstantSymbol[M2, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{M}\),\(\*OverscriptBox[\(\kappa\),\((2)\)]\)],\((2)\)]\)");
```

```
DefConstantSymbol[M3, PrintAs →
  "\!\(\*SubscriptBox[\(\mathcal{M}\),\(\*OverscriptBox[\(\kappa\),\((2)\)]\)],\((3)\)]\)");
```

$$\begin{aligned}
& \kappa_1^{(2)} \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} + \kappa_2^{(2)} \mathcal{K}^{\alpha\beta\chi} \mathcal{K}_{\beta\alpha\chi} + \kappa_3^{(2)} \mathcal{K}^{\alpha\beta} \mathcal{K}^{\chi}_{\beta\chi} + \kappa_1^{(4)} \partial_{\beta} \mathcal{K}^{\delta}_{\chi\delta} \partial^{\chi} \mathcal{K}^{\alpha\beta}_{\alpha} + \kappa_2^{(4)} \partial_{\chi} \mathcal{K}^{\delta}_{\beta\delta} \partial^{\chi} \mathcal{K}^{\alpha\beta}_{\alpha} + \\
& \kappa_3^{(4)} \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}_{\alpha\chi} + \kappa_4^{(4)} \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}_{\beta\chi} + \kappa_5^{(4)} \partial^{\chi} \mathcal{K}^{\alpha\beta}_{\alpha} \partial_{\delta} \mathcal{K}^{\delta}_{\beta\chi} + \kappa_6^{(4)} \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}_{\chi\alpha} + \\
& \kappa_7^{(4)} \partial^{\chi} \mathcal{K}^{\alpha\beta}_{\alpha} \partial_{\delta} \mathcal{K}^{\delta}_{\chi\beta} + \kappa_9^{(4)} \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}_{\beta\chi} + \kappa_{15}^{(4)} \partial_{\delta} \mathcal{K}_{\alpha\beta\chi} \partial^{\delta} \mathcal{K}^{\alpha\beta\chi} + \kappa_{16}^{(4)} \partial_{\delta} \mathcal{K}_{\beta\alpha\chi} \partial^{\delta} \mathcal{K}^{\alpha\beta\chi}
\end{aligned} \tag{1}$$

Bifurcation

$$\begin{aligned}
& \kappa_1^{(2)} \mathcal{K}^{\alpha\beta\chi} \left(\mathcal{K}_{\alpha\beta\chi} + \mathcal{K}_{\beta\alpha\chi} \right) + \kappa_3^{(2)} \mathcal{K}^{\alpha\beta} \mathcal{K}^{\chi}_{\beta\chi} + \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \left(\kappa_3^{(4)} \partial_{\delta} \mathcal{K}^{\delta}_{\alpha\chi} + \kappa_6^{(4)} \partial_{\delta} \mathcal{K}^{\delta}_{\chi\alpha} \right) + \\
& \partial^{\chi} \mathcal{K}^{\alpha\beta}_{\alpha} \left(\kappa_1^{(4)} \partial_{\beta} \mathcal{K}^{\delta}_{\chi\delta} + \kappa_2^{(4)} \partial_{\chi} \mathcal{K}^{\delta}_{\beta\delta} + \kappa_5^{(4)} \partial_{\delta} \mathcal{K}^{\delta}_{\beta\chi} + \kappa_7^{(4)} \partial_{\delta} \mathcal{K}^{\delta}_{\chi\beta} \right) + \\
& \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \left(\kappa_4^{(4)} \partial_{\delta} \mathcal{K}^{\delta}_{\beta\chi} + \kappa_9^{(4)} \partial_{\delta} \mathcal{K}^{\delta}_{\beta\chi} \right) + \kappa_{15}^{(4)} \left(\partial_{\delta} \mathcal{K}_{\alpha\beta\chi} + \partial_{\delta} \mathcal{K}_{\beta\alpha\chi} \right) \partial^{\delta} \mathcal{K}^{\alpha\beta\chi}
\end{aligned} \tag{2}$$

```

ParticleSpectrum[M1*kNS[a, b, c]*(kNS[-a, -b, -c] + kNS[-b, -a, -c]) +
  M3*kNS[a, -a, b]*kNS[c, -b, -c] + CD[-b][kNS[a, b, c]]*
  (K3*CD[-d][kNS[-a, -c, d]] + K6*CD[-d][kNS[-c, -a, d]]) +
  CD[c][kNS[a, -a, b]]*(K1*CD[-b][kNS[d, -c, -d]] +
  K2*CD[-c][kNS[d, -b, -d]] + K5*CD[-d][kNS[-b, -c, d]] +
  K7*CD[-d][kNS[-c, -b, d]]) + CD[-a][kNS[a, b, c]]*
  (K4*CD[-d][kNS[-b, -c, d]] + K9*CD[-d][kNS[d, -b, -c]]) +
  K15*(CD[-d][kNS[-a, -b, -c]] + CD[-d][kNS[-b, -a, -c]])*
  CD[d][kNS[a, b, c]],
TheoryName → "R1B1", Method → "Hard", MaxLaurentDepth → 1];

```

⋮ **Values:** The argument InputallCases is not a valid Association or a list of rules.

⋮ **Join:** Nonatomic expression expected at position 1 in Join[InputallCases]. ⓘ

$$\begin{aligned}
& \mathcal{K}^{\alpha\beta\chi} \left(\binom{2}{K_1} \mathcal{K}_{\alpha\beta\chi} + \binom{2}{K_2} \mathcal{K}_{\beta\alpha\chi} \right) + \binom{2}{K_3} \mathcal{K}_{\alpha}^{\alpha}{}^{\beta} \mathcal{K}_{\beta\chi}^{\chi} + \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \left(\binom{4}{K_3} \partial_{\delta} \mathcal{K}_{\alpha\chi}^{\delta} + \binom{4}{K_6} \partial_{\delta} \mathcal{K}_{\chi\alpha}^{\delta} \right) + \\
& \partial^{\chi} \mathcal{K}_{\alpha}^{\alpha}{}^{\beta} \left(\binom{4}{K_1} \partial_{\beta} \mathcal{K}_{\chi\delta}^{\delta} + \binom{4}{K_2} \partial_{\chi} \mathcal{K}_{\beta\delta}^{\delta} + \binom{4}{K_5} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} + \binom{4}{K_7} \partial_{\delta} \mathcal{K}_{\chi\beta}^{\delta} \right) + \\
& \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \left(\binom{4}{K_4} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} + \binom{4}{K_9} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} \right) + \left(\binom{4}{K_{15}} \partial_{\delta} \mathcal{K}_{\alpha\beta\chi}^{\delta} + \binom{4}{K_{16}} \partial_{\delta} \mathcal{K}_{\beta\alpha\chi}^{\delta} \right) \partial^{\delta} \mathcal{K}^{\alpha\beta\chi}
\end{aligned} \tag{3}$$

```

ParticleSpectrum[kNS[a, b, c]*(M1*kNS[-a, -b, -c] + M2*kNS[-b, -a, -c]) +
  M3*kNS[a, -a, b]*kNS[c, -b, -c] + CD[-b][kNS[a, b, c]]*
  (K3*CD[-d][kNS[-a, -c, d]] + K6*CD[-d][kNS[-c, -a, d]]) +
  CD[c][kNS[a, -a, b]]*(K1*CD[-b][kNS[d, -c, -d]] +
  K2*CD[-c][kNS[d, -b, -d]] + K5*CD[-d][kNS[-b, -c, d]] +
  K7*CD[-d][kNS[-c, -b, d]]) + CD[-a][kNS[a, b, c]]*
  (K4*CD[-d][kNS[-b, -c, d]] + K9*CD[-d][kNS[d, -b, -c]]) +
  (K15*CD[-d][kNS[-a, -b, -c]] + K16*CD[-d][kNS[-b, -a, -c]])*
  CD[d][kNS[a, b, c]],
TheoryName → "R1B2", Method → "Hard", MaxLaurentDepth → 1];

```

⋮ **Values:** The argument InputallCases is not a valid Association or a list of rules.

⋮ **Join:** Nonatomic expression expected at position 1 in Join[InputallCases]. ⓘ

$$\begin{aligned}
& \mathcal{K}^{\alpha\beta\chi} \left(\binom{2}{K_1} \mathcal{K}_{\alpha\beta\chi} + \binom{2}{K_2} \mathcal{K}_{\beta\alpha\chi} \right) + \binom{2}{K_3} \mathcal{K}_{\alpha}^{\alpha}{}^{\beta} \mathcal{K}_{\beta\chi}^{\chi} + \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \left(\binom{4}{K_3} \partial_{\delta} \mathcal{K}_{\alpha\chi}^{\delta} + \binom{4}{K_6} \partial_{\delta} \mathcal{K}_{\chi\alpha}^{\delta} \right) + \\
& \partial^{\chi} \mathcal{K}_{\alpha}^{\alpha}{}^{\beta} \left(\binom{4}{K_1} \partial_{\beta} \mathcal{K}_{\chi\delta}^{\delta} + \binom{4}{K_2} \partial_{\chi} \mathcal{K}_{\beta\delta}^{\delta} + \binom{4}{K_5} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} + \binom{4}{K_7} \partial_{\delta} \mathcal{K}_{\chi\beta}^{\delta} \right) + \\
& \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \left(\binom{4}{K_4} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} + \binom{4}{K_9} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} \right) + \left(\binom{4}{K_{15}} \partial_{\delta} \mathcal{K}_{\alpha\beta\chi}^{\delta} + \binom{4}{K_{16}} \partial_{\delta} \mathcal{K}_{\beta\alpha\chi}^{\delta} \right) \partial^{\delta} \mathcal{K}^{\alpha\beta\chi}
\end{aligned} \tag{4}$$

```

ParticleSpectrum[kNS[a, b, c]*(M1*kNS[-a, -b, -c] + M2*kNS[-b, -a, -c]) +
  M3*kNS[a, -a, b]*kNS[c, -b, -c] + CD[-b][kNS[a, b, c]]*
  (K3*CD[-d][kNS[-a, -c, d]] + K6*CD[-d][kNS[-c, -a, d]]) +
  CD[c][kNS[a, -a, b]]*(K1*CD[-b][kNS[d, -c, -d]] +
    K2*CD[-c][kNS[d, -b, -d]] + K5*CD[-d][kNS[-b, -c, d]] +
    K7*CD[-d][kNS[-c, -b, d]]) + CD[-a][kNS[a, b, c]]*
  (K4*CD[-d][kNS[-b, -c, d]] + K9*CD[-d][kNS[d, -b, -c]]) +
  (K15*CD[-d][kNS[-a, -b, -c]] + K16*CD[-d][kNS[-b, -a, -c]])*
  CD[d][kNS[a, b, c]],
TheoryName → "R1B3", Method → "Hard", MaxLaurentDepth → 1];

```

... **Values**: The argument InputallCases is not a valid Association or a list of rules.

... **General**: Further output of Values::invr1 will be suppressed during this calculation. ⓘ

... **Join**: Nonatomic expression expected at position 1 in Join[InputallCases]. ⓘ

... **General**: Further output of Join::normal will be suppressed during this calculation. ⓘ

$$\begin{aligned}
& \mathcal{K}^{\alpha\beta\chi} \left(\binom{2}{\kappa_1} \mathcal{K}_{\alpha\beta\chi} + \binom{2}{\kappa_2} \mathcal{K}_{\beta\alpha\chi} \right) + \binom{2}{\kappa_3} \mathcal{K}^{\alpha\beta} \mathcal{K}^{\chi}_{\beta\chi} + \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \left(\binom{4}{\kappa_3} \partial_{\delta} \mathcal{K}_{\alpha\chi}^{\delta} + \binom{4}{\kappa_6} \partial_{\delta} \mathcal{K}_{\chi\alpha}^{\delta} \right) + \\
& \partial^{\chi} \mathcal{K}^{\alpha\beta}_{\alpha} \left(\binom{4}{\kappa_1} \partial_{\beta} \mathcal{K}_{\chi\delta}^{\delta} + \binom{4}{\kappa_2} \partial_{\chi} \mathcal{K}_{\beta\delta}^{\delta} + \binom{4}{\kappa_5} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} + \binom{4}{\kappa_7} \partial_{\delta} \mathcal{K}_{\chi\beta}^{\delta} \right) + \\
& \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \left(\binom{4}{\kappa_4} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} + \binom{4}{\kappa_9} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} \right) + \left(\binom{4}{\kappa_{15}} \partial_{\delta} \mathcal{K}_{\alpha\beta\chi} + \binom{4}{\kappa_{16}} \partial_{\delta} \mathcal{K}_{\beta\alpha\chi} \right) \partial^{\delta} \mathcal{K}^{\alpha\beta\chi}
\end{aligned} \tag{5}$$

```

ParticleSpectrum[kNS[a, b, c]*(M1*kNS[-a, -b, -c] + M2*kNS[-b, -a, -c]) +
  M3*kNS[a, -a, b]*kNS[c, -b, -c] + CD[-b][kNS[a, b, c]]*
  (K3*CD[-d][kNS[-a, -c, d]] + K6*CD[-d][kNS[-c, -a, d]]) +
  CD[c][kNS[a, -a, b]]*(K1*CD[-b][kNS[d, -c, -d]] +
    K2*CD[-c][kNS[d, -b, -d]] + K5*CD[-d][kNS[-b, -c, d]] +
    K7*CD[-d][kNS[-c, -b, d]]) + CD[-a][kNS[a, b, c]]*
  (K4*CD[-d][kNS[-b, -c, d]] + K9*CD[-d][kNS[d, -b, -c]]) +
  (K15*CD[-d][kNS[-a, -b, -c]] + K16*CD[-d][kNS[-b, -a, -c]])*
  CD[d][kNS[a, b, c]],
TheoryName → "R1B4", Method → "Hard", MaxLaurentDepth → 1];

```

$$\begin{aligned}
& \mathcal{K}^{\alpha\beta\chi} \left(\binom{2}{\kappa_1} \mathcal{K}_{\alpha\beta\chi} + \binom{2}{\kappa_2} \mathcal{K}_{\beta\alpha\chi} \right) + \binom{2}{\kappa_3} \mathcal{K}^{\alpha\beta} \mathcal{K}^{\chi}_{\beta\chi} + \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \left(\binom{4}{\kappa_3} \partial_{\delta} \mathcal{K}_{\alpha\chi}^{\delta} + \binom{4}{\kappa_6} \partial_{\delta} \mathcal{K}_{\chi\alpha}^{\delta} \right) + \\
& \partial^{\chi} \mathcal{K}^{\alpha\beta}_{\alpha} \left(\binom{4}{\kappa_1} \partial_{\beta} \mathcal{K}_{\chi\delta}^{\delta} + \binom{4}{\kappa_2} \partial_{\chi} \mathcal{K}_{\beta\delta}^{\delta} + \binom{4}{\kappa_5} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} + \binom{4}{\kappa_7} \partial_{\delta} \mathcal{K}_{\chi\beta}^{\delta} \right) + \\
& \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \left(\binom{4}{\kappa_4} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} + \binom{4}{\kappa_9} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} \right) + \left(\binom{4}{\kappa_{15}} \partial_{\delta} \mathcal{K}_{\alpha\beta\chi} + \binom{4}{\kappa_{16}} \partial_{\delta} \mathcal{K}_{\beta\alpha\chi} \right) \partial^{\delta} \mathcal{K}^{\alpha\beta\chi}
\end{aligned} \tag{6}$$

```

ParticleSpectrum[kNS[a, b, c] * (M1 * kNS[-a, -b, -c] + M2 * kNS[-b, -a, -c]) +
  M3 * kNS[a, -a, b] * kNS[c, -b, -c] + CD[-b][kNS[a, b, c]] *
  (K3 * CD[-d][kNS[-a, -c, d]] + K6 * CD[-d][kNS[-c, -a, d]]) +
  CD[c][kNS[a, -a, b]] * (K1 * CD[-b][kNS[d, -c, -d]] +
    K2 * CD[-c][kNS[d, -b, -d]] + K5 * CD[-d][kNS[-b, -c, d]] +
    K7 * CD[-d][kNS[-c, -b, d]]) + CD[-a][kNS[a, b, c]] *
  (K4 * CD[-d][kNS[-b, -c, d]] + K9 * CD[-d][kNS[d, -b, -c]]) +
  (K15 * CD[-d][kNS[-a, -b, -c]] + K16 * CD[-d][kNS[-b, -a, -c]]) *
  CD[d][kNS[a, b, c]],
TheoryName → "R1C1", Method → "Hard", MaxLaurentDepth → 1];

```

$$\begin{aligned}
& \mathcal{K}^{\alpha\beta\chi} \left(\binom{2}{K_1} \mathcal{K}_{\alpha\beta\chi} + \binom{2}{K_2} \mathcal{K}_{\beta\alpha\chi} \right) + \binom{2}{K_3} \mathcal{K}^{\alpha\beta} \mathcal{K}^{\chi}_{\beta\chi} + \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \left(\binom{4}{K_3} \partial_{\delta} \mathcal{K}_{\alpha\chi}^{\delta} + \binom{4}{K_6} \partial_{\delta} \mathcal{K}_{\chi\alpha}^{\delta} \right) + \\
& \partial^{\chi} \mathcal{K}^{\alpha\beta}_{\alpha} \left(\binom{4}{K_1} \partial_{\beta} \mathcal{K}_{\chi\delta}^{\delta} + \binom{4}{K_2} \partial_{\chi} \mathcal{K}_{\beta\delta}^{\delta} + \binom{4}{K_5} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} + \binom{4}{K_7} \partial_{\delta} \mathcal{K}_{\chi\beta}^{\delta} \right) + \\
& \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \left(\binom{4}{K_4} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} + \binom{4}{K_9} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} \right) + \left(\binom{4}{K_{15}} \partial_{\delta} \mathcal{K}_{\alpha\beta\chi} + \binom{4}{K_{16}} \partial_{\delta} \mathcal{K}_{\beta\alpha\chi} \right) \partial^{\delta} \mathcal{K}^{\alpha\beta\chi}
\end{aligned} \tag{7}$$

```

ParticleSpectrum[kNS[a, b, c] * (M1 * kNS[-a, -b, -c] + M2 * kNS[-b, -a, -c]) +
  M3 * kNS[a, -a, b] * kNS[c, -b, -c] + CD[-b][kNS[a, b, c]] *
  (K3 * CD[-d][kNS[-a, -c, d]] + K6 * CD[-d][kNS[-c, -a, d]]) +
  CD[c][kNS[a, -a, b]] * (K1 * CD[-b][kNS[d, -c, -d]] +
    K2 * CD[-c][kNS[d, -b, -d]] + K5 * CD[-d][kNS[-b, -c, d]] +
    K7 * CD[-d][kNS[-c, -b, d]]) + CD[-a][kNS[a, b, c]] *
  (K4 * CD[-d][kNS[-b, -c, d]] + K9 * CD[-d][kNS[d, -b, -c]]) +
  (K15 * CD[-d][kNS[-a, -b, -c]] + K16 * CD[-d][kNS[-b, -a, -c]]) *
  CD[d][kNS[a, b, c]],
TheoryName → "R1C2", Method → "Hard", MaxLaurentDepth → 1];

```

$$\begin{aligned}
& \mathcal{K}^{\alpha\beta\chi} \left(\binom{2}{K_1} \mathcal{K}_{\alpha\beta\chi} + \binom{2}{K_2} \mathcal{K}_{\beta\alpha\chi} \right) + \binom{2}{K_3} \mathcal{K}^{\alpha\beta} \mathcal{K}^{\chi}_{\beta\chi} + \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \left(\binom{4}{K_3} \partial_{\delta} \mathcal{K}_{\alpha\chi}^{\delta} + \binom{4}{K_6} \partial_{\delta} \mathcal{K}_{\chi\alpha}^{\delta} \right) + \\
& \partial^{\chi} \mathcal{K}^{\alpha\beta}_{\alpha} \left(\binom{4}{K_1} \partial_{\beta} \mathcal{K}_{\chi\delta}^{\delta} + \binom{4}{K_2} \partial_{\chi} \mathcal{K}_{\beta\delta}^{\delta} + \binom{4}{K_5} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} + \binom{4}{K_7} \partial_{\delta} \mathcal{K}_{\chi\beta}^{\delta} \right) + \\
& \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \left(\binom{4}{K_4} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} + \binom{4}{K_9} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} \right) + \left(\binom{4}{K_{15}} \partial_{\delta} \mathcal{K}_{\alpha\beta\chi} + \binom{4}{K_{16}} \partial_{\delta} \mathcal{K}_{\beta\alpha\chi} \right) \partial^{\delta} \mathcal{K}^{\alpha\beta\chi}
\end{aligned} \tag{8}$$


```

ParticleSpectrum[kNS[a, b, c] * (M1 * kNS[-a, -b, -c] + M2 * kNS[-b, -a, -c]) +
  M3 * kNS[a, -a, b] * kNS[c, -b, -c] + CD[-b][kNS[a, b, c]] *
  (K3 * CD[-d][kNS[-a, -c, d]] + K6 * CD[-d][kNS[-c, -a, d]]) +
  CD[c][kNS[a, -a, b]] * (K1 * CD[-b][kNS[d, -c, -d]] +
    K2 * CD[-c][kNS[d, -b, -d]] + K5 * CD[-d][kNS[-b, -c, d]] +
    K7 * CD[-d][kNS[-c, -b, d]]) + CD[-a][kNS[a, b, c]] *
  (K4 * CD[-d][kNS[-b, -c, d]] + K9 * CD[-d][kNS[d, -b, -c]]) +
  (K15 * CD[-d][kNS[-a, -b, -c]] + K16 * CD[-d][kNS[-b, -a, -c]]) *
  CD[d][kNS[a, b, c]],
TheoryName → "R1C3", Method → "Hard", MaxLaurentDepth → 1];

```

$$\begin{aligned}
& \mathcal{K}^{\alpha\beta\chi} \left(\binom{(2)}{K_1} \mathcal{K}_{\alpha\beta\chi} + \binom{(2)}{K_2} \mathcal{K}_{\beta\alpha\chi} \right) + \binom{(2)}{K_3} \mathcal{K}^{\alpha\beta} \mathcal{K}^{\chi}_{\beta\chi} + \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \left(\binom{(4)}{K_3} \partial_{\delta} \mathcal{K}^{\delta}_{\alpha\chi} + \binom{(4)}{K_6} \partial_{\delta} \mathcal{K}^{\delta}_{\chi\alpha} \right) + \\
& \partial^{\chi} \mathcal{K}^{\alpha\beta}_{\alpha} \left(\binom{(4)}{K_1} \partial_{\beta} \mathcal{K}^{\delta}_{\chi\delta} + \binom{(4)}{K_2} \partial_{\chi} \mathcal{K}^{\delta}_{\beta\delta} + \binom{(4)}{K_5} \partial_{\delta} \mathcal{K}^{\delta}_{\beta\chi} + \binom{(4)}{K_7} \partial_{\delta} \mathcal{K}^{\delta}_{\chi\beta} \right) + \\
& \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \left(\binom{(4)}{K_4} \partial_{\delta} \mathcal{K}^{\delta}_{\beta\chi} + \binom{(4)}{K_9} \partial_{\delta} \mathcal{K}^{\delta}_{\beta\chi} \right) + \left(\binom{(4)}{K_{15}} \partial_{\delta} \mathcal{K}^{\delta}_{\alpha\beta\chi} + \binom{(4)}{K_{16}} \partial_{\delta} \mathcal{K}^{\delta}_{\beta\alpha\chi} \right) \partial^{\delta} \mathcal{K}^{\alpha\beta\chi}
\end{aligned} \tag{9}$$

```

ParticleSpectrum[kNS[a, b, c] * (M1 * kNS[-a, -b, -c] + M2 * kNS[-b, -a, -c]) +
  M3 * kNS[a, -a, b] * kNS[c, -b, -c] + CD[-b][kNS[a, b, c]] *
  (K3 * CD[-d][kNS[-a, -c, d]] + K6 * CD[-d][kNS[-c, -a, d]]) +
  CD[c][kNS[a, -a, b]] * (K1 * CD[-b][kNS[d, -c, -d]] +
    K2 * CD[-c][kNS[d, -b, -d]] + K5 * CD[-d][kNS[-b, -c, d]] +
    K7 * CD[-d][kNS[-c, -b, d]]) + CD[-a][kNS[a, b, c]] *
  (K4 * CD[-d][kNS[-b, -c, d]] + K9 * CD[-d][kNS[d, -b, -c]]) +
  (K15 * CD[-d][kNS[-a, -b, -c]] + K16 * CD[-d][kNS[-b, -a, -c]]) *
  CD[d][kNS[a, b, c]],
TheoryName → "R1C4", Method → "Hard", MaxLaurentDepth → 1];

```